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Status-Aspirational Pricing: The "Chivas Regal" Strategy in U.S. Higher Education, 2006–2012

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Abstract

This paper examines the effect of status loss on organizations' price-setting behavior. We predict, counter to current status theory and aligned with performance feedback theory, that a status decline prompts certain organizations to charge higher prices and that there are two kinds of organizations most prone to make such price increases: those with broad appeal across disconnected types of customers and those whose most strategically similar rivals have charged high prices previously. Using panel data from *U.S. News & World Report*'s annual rankings of private colleges and universities from 2005 to 2012, we model the effect of drops in rank that take a school below an aspiration level. We find that schools set tuition higher after a sharp decline in rank, particularly those that appeal widely to college applicants and whose rivals are relatively more expensive. This study presents a dynamic conception of status that differs from the prevailing view of status as a stable asset that yields concrete benefits. In contrast to past work that has assumed that organizations passively experience negative effects when their status falls, our results show that organizations actively respond to status loss. Status is a performance-related goal for such producers, who may increase prices as they work to recover lost ground after a status decline.

Keywords: status, performance feedback theory, decision making, college rankings, Chivas Regal effect, pricing, signaling

Whether status is seen as a means of detecting otherwise obscured quality or as an end in itself, a core insight of previous research in organization theory is that high-status organizations command higher prices than low-status organizations for their goods and services. Benjamin and Podolny (1999) found that elite wineries enjoy higher prices for their wines than do their lower-status rivals; Stuart, Hoang, and Hybels (1999) established that biotech ventures with more

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prestigious affiliates enjoy better initial valuations; and Uzzi and Lancaster (2004) identified a strong positive effect of corporate law firm status on the price of legal services. What remains less clear, however, is the impact that an organization's changes in status have on its price-setting behavior. Specifically, what is the effect that a loss of status has on the price of an organization's goods and services?

Researchers have given little attention to how an organization responds after its status falls. Focusing explicitly on these behavioral responses is important because it is likely that ''assuming away'' agency (Burt, 2010: 222) has created an incomplete view of how status shapes outcomes in markets. By agency, we mean action taken in the face of expectations imposed by other market actors (see Jensen, Kim, and Kim, 2011). Prevailing status theory (e.g., Podolny, 2005; Sauder, Lynn, and Podolny, 2012) suggests that when an organization (or individual) slips in status, status loss leads to lower rates of innovation, higher costs, elevated exit chances, lower growth rates, and lower prices. Seen through this theoretical lens, status is strongly deterministic, and the implicit picture is clear: when an organization's status falls, it passively suffers a future performance decline with low chances for recovery.

An emphatically more agentic conception of status dynamics in markets is necessary. Focusing on active responses to the pain of status loss is important for producing better models of status-based performance (measured by organizational growth, survival, or profitability) and of industry-wide status distributions (Sauder, Lynn, and Podolny, 2012: 275). If we take into account the triggering effect that status loss can exert—or the satiating effect a status gain can have—we may find that status shapes both organizational performance and global status distributions in ways that differ from what prevailing wisdom predicts. The core issue is that a loss in status elicits efforts to recover, which can curb the unfavorable effects of these declines on organizational performance, while also shaking up and ultimately tightening status distributions otherwise thought to fan out irreversibly with time (cf. Merton, 1968; Askin, Bothner, and Lee, 2015).

To develop an agentic theory of the effect of status loss on future pricing, we draw extensively on performance feedback theory (Greve, 2003; Baum et al., 2005; Shipilov, Li, and Greve, 2011), which is not overly deterministic and gives significant weight to agency. It posits that an organization receives feedback from its external environment, such as changes to its rank in a dynamic contest for status, and adjusts its conduct, such as its price-setting behavior, if that feedback indicates it has fallen short of its aspirations. Using performance feedback theory as a basis for advancing status theory also requires us to view status and price differently from how they have been viewed in prior work on organizational status.

Consistent with signaling theory (Spence, 1974) and hedonic pricing models (Rosen, 1974; Feenstra, 1995), Benjamin and Podolny (1999) in their study of wine and Uzzi and Lancaster (2004) in their study of legal services conceived of status as a largely stable trait that buyers desire: consumers of wine take winery status as a signal of difficult-to-observe quality, and corporate clients derive satisfaction from the status of the law firms whose services they buy. Price is then a natural reflection of consumer demand: buyers consider a vertical array of outputs, differentiated by producer status, and are more willing to pay higher prices for the goods or services of higher-status organizations.

Using between-organization models, Benjamin and Podolny (1999) and Uzzi and Lancaster (2004) found positive effects of organizational status on price. Inspired by the focus on dynamics in performance feedback theory, we instead view status as a mutable position in a tournament in which producers seek to rank favorably (see also Sauder and Lancaster, 2006; Espeland and Sauder, 2007). Status is therefore not exclusively a trait that buyers value; it is also a salient, performance-related goal for producers.

When a producer's status falls below its aspirations, it may use price as an ''adaptive device'' (Cyert and March, 1963: 152) in an effort to recover lost ground. We thus cast price setting as an organizational decision based on status relative to aspirations, and we discuss organizations as prone to increase prices after a drop in status in order to send a quality signal conducive to status recovery. In this sense, we anchor on a different side of the ''dual role of price'' (Völkner and Hofmann, 2007: 194). Following Leavitt's (1954) early distinction, we focus on price not simply as a ''sacrifice'' for consumers but also as a ''symbol'' used strategically by status-aspiring organizations. Several other studies have drawn a similar distinction. For instance, Rao and Monroe (1988: 253) distinguished between price as an ''indicator of sacrifice'' in economic theory and price as a ''quality cue'' in behavioral research, and Fruchter (2009: 479) distinguished between price ''as a monetary constraint'' and price ''as a signal of quality.'' Consistent with this second view of the role of price, and using within- rather than between-organization models, we test the hypothesis that an organization's loss in status, not its gain, triggers a price increase.

Our departures from prior theory are not necessary, however, for understanding the consequences of status in all empirical settings. One can certainly imagine markets in which virtually all variance in producers' status is time-invariant and prices are primarily determined by buyers' preferences. In such cases, the traditional conception of status is the correct theoretical lens: status acts as a signal or a good that raises buyers' willingness to pay. Yet in many contexts, whose main features we later sketch as scope conditions, producers do see status as a performance dimension (Sauder and Espeland, 2009) on which they vary over time, rendering relatively passive responses to status decline unlikely. For these settings, it is necessary to reconsider the nature of the effect of status on price and the possibility of a producer taking action in response to a decline in status.

RAISING PRICES IN REACTION TO STATUS LOSS

To illustrate our main hypothesis, we have constructed a theoretically informed narrative (cf. Greve, 2003: 53–60) centered on a focal producer *A*, an organization occupying a rank in a dynamic contest for status. *A* sells a good, in competition with other producers in the set ***A***, to a group of buyers ***B***. Both groups—***A*** and ***B***—place a high value on the status levels of the producers and their goods in the marketplace: higher-status members of ***A*** attract status-seeking members of ***B***, who in turn intrinsically value their association with higher-status producers and their wares. *A*'s decision makers are therefore compelled to take steps to ensure the continual maintenance or improvement of their organizational status: it is a goal unto itself. In addition, in the simple account that follows, *A* takes actions that clearly not all producers across all industries

can take, but before describing the factors that bracket the generality of these actions, we lay out their antecedents and expected consequences.

Suppose that, in the current iteration of the contest, *A* observes that it has slipped sharply in status, and this decline means that *A* is now significantly shy of its aspirational status level. Within the status hierarchy, this aspiration level is a narrow region in which *A* is comfortable—a band in the status order toward which *A* is ''psychologically neutral'' (Kameda and Davis, 1990: 56; Greve, 2003: 39). Because *A* has fallen below this band, *A* is no longer satisfied but rather suffers the pain of status loss. In line with prospect theory (Kahneman and Tversky, 1979) and with Burt's (1982, 2010) model of peer effects, this displeasure is greater than the pleasure *A* would enjoy if it had instead ascended above this band.

Two organizational changes result from *A*'s dissatisfaction. First, as prospect theory suggests, *A* is now more willing to engage in risky behavior (Kahneman and Tversky, 1979, 1984; Bothner, Kang, and Stuart, 2007). Second, as performance feedback theory indicates, *A*'s sharp distaste for status loss evokes deliberations on possible strategies for recapturing lost ground. We envision *A*'s deliberations as a special form of problemistic search triggered by unmet objectives (Cyert and March, 1963; Shipilov, Li, and Greve, 2011). These deliberations are related directly to price setting and are separate from ongoing, institutionalized mechanisms for status maintenance (cf. Greve, 2003: 54–56), such as quality investments, strategic gift-giving, and endorsement-seeking (Podolny and Phillips, 1996; Stuart, Hoang, and Hybels, 1999).

Focusing here on *A*'s strategic search in response to status loss, imagine that *A* considers the possibility of raising the price of its good. Doing so is, in effect, acting on a folk theorem known as the Chivas Regal strategy. Named after a previously middling brand of whiskey purported to have doubled its sales by doubling its price (Finkelmeyer, 2011), the Chivas Regal strategy is often described as an effort to enhance status by price signaling: raising price is a way for the focal organization to increase its perceived quality and associate more closely with its more-elite peers. When a producer takes this approach, it is choosing a price it believes will maximize its subjective function of status and revenue: the organization attempts to send a quality signal, but not with a price so high that prospective buyers recoil.

The Chivas Regal strategy is consistent with a long history of work emphasizing the symbolic role of prices. Leavitt (1954: 206) was among the first to stress that price could serve as a symbol of prestige, arguing that a high price may attract instead of repel and citing as an example the ''phenomenon of the dormant shirt that comes to life when the price is marked *up*.'' In contemporary experimental work, Plassmann and her colleagues (2008) established that when individuals are (falsely) informed that a given wine commands a higher price than a physically identical wine, they give the allegedly more expensive wine a better rating. Perhaps most succinctly demonstrating the potent impact of price on perceptions, Espay and colleagues (2015) found that placebos described as ''expensive'' are more effective than those described as ''cheap'' in the treatment of Parkinson's disease.

Numerous studies similarly suggest that price and perceived quality are often positively associated when comparing organizations or their products at a single point in time. For example, although the magnitude of the correlation between price and quality varies across markets, meta-analyses by Tellis and

Wernerfelt (1987), Rao and Monroe (1988), and Völckner and Hofmann (2007) are all consistent with the axiom ''you get what you pay for.'' Although we do not cast our focal producer as explicitly aware of particular, stylized facts reported in the scientific literature, *A* is not so boundedly rational as to be unaware of general patterns relevant to its goal of status recovery. Consistent with a (likely rough) sense of these patterns, *A* may also choose to raise price because it believes that this will improve how it is perceived among those shaping its status.

A's assessment of the gap between its current and desired status elicits further deliberation and ultimately a decision. Separate from the problemistic search caused by status loss, the focal producer must of course also consider an array of other factors as it settles on the price of its good, including the same cost and demand drivers it routinely considers when making pricing decisions. *A* must finally make a judgment about whether raising its price makes sense in light of the risks involved: demand for *A*'s good may suffer and, in some contexts, the composition of its buyers may grow less heterogeneous, or *A* may be stigmatized for violating fairness norms on pricing (Kahneman, Knetsch, and Thaler, 1986). One can thus liken the pricing decision-making process to titration, the practice in medicine of routinely measuring and adjusting the dose of a medication administered in order to maximize its benefits and minimize its side effects. The status-aspiring organization dances around a ''fair'' price that is close enough to an appropriate reflection of its current status and yet still sends a signal conducive to the status to which it aspires.

More-complex considerations surround how certain *A* is that a higher price will translate into status recovery, including aspects of the organization's position in its market and its tolerance for risk. Suppose, for now, however, that *A*'s market position and its broader institutional context are such that the causal sequence we have sketched ends with *A*'s decision to engage in status-aspirational pricing: *A*'s status drops below its aspiration level; the pain of this deviation evokes problemistic deliberation as well as a greater willingness to take risks; and *A*, aiming to recover in status, decides that raising its price is the right decision despite the risk involved. Our main hypothesis is thus:

Hypothesis 1: A focal organization increases its price in reaction to status loss.

Flexibility in Price Setting

There are two circumstances under which organizations are most prone to react to status loss by raising their prices. The first has to do with risk taking and an organization's consumer-conferred freedom to charge more, the second with the freedom afforded by its competitors in the market. Following from our discussion of price hikes as cases of organizational risk taking, we expect the focal producer *A* to react most strongly to a status decline if *A* appeals broadly to disparate kinds of buyers (cf. Burt, 1992: fig. 6.6). Imagine that buyers ***B*** are differentiated in their preferences for certain producers over others. Such a pattern might emerge as the result of many producers' tendencies to target specific locations in a technical, geographical, or product-features space and then target particular kinds of buyers similarly distributed across that space (cf. White, 2002: 107–108). Imagine that, by contrast, *A* appeals to a broad array of these buyers rather than selling just to a narrow subset. Suppose further that

the structural features of *A*'s market are such that breadth of appeal, rather than specialization (e.g., Carroll and Swaminathan, 2000), raises the attractiveness and durability of producers' positions in that market (e.g., White, 1981: 540–541; Stevens, 2007). In such a setting, our reasons for anticipating that this breadth of appeal will magnify the strength of *A*'s reaction to a status decline are twofold: the first is purely structural; the second is perceptual and more intriguing theoretically (see Podolny, 2001, on ''pipes'' versus ''prisms'' for an analogous distinction).

First, the risks often associated with price increases are lower for an organization that enjoys broad appeal: such a producer is aware of its options and is appropriately hedged should some consumers defect. This claim follows several long-standing theoretical approaches, which include resource dependence (Pfeffer and Salancik, 1978), ecological portrayals of niche width as a buffer against uncertainty (Hannan and Freeman, 1989; Péli, 1997), and structural hole theory, which highlights the freedom to experiment enjoyed by occupants of brokerage positions (Burt, 1992: 195–197). These research traditions, which stress the advantages of being minimally dependent on any one exchange partner, jointly imply that organizations exhibiting wide, diverse appeal will also deem decisions that are typically seen as high risk as less precarious.

Second, an organization with broad appeal occupies a subjectively advantageous position within its market. Breadth of appeal is associated with an organization residing near the core (and away from peripheral players) of its market, with connections extending to multiple groups. Such reach makes an organization appear more credible than its more resource-dependent rivals on the periphery. Here we again diverge from research underscoring the identity-related advantages of focus and specialization (e.g., Zuckerman, 1999, 2000; Leahey, 2007; Leahey and Reikowsky, 2008), instead focusing on the confirmation of credibility provided when normally disparate or disagreeing parties reach consensus on a focal individual or organization. We view producer–buyer ties as implicit endorsements that are especially valuable for the producer if its buyers are diffuse and otherwise non-overlapping in their choice-sets. This intuition is perhaps best illustrated by an example at the individual level: consider a job candidate who is able to earn praise from several members of a hiring committee who are typically at each other's throats. Getting broad support from those with divergent tastes signals that the candidate, more than his or her narrowly supported rivals, ''is worthy'' (cf. Centola and Macy, 2007). Research on the performance-related advantages of betweenness centrality (Rowley et al., 2004) and robust positions (Bothner, Smith, and White, 2010) in interorganizational networks is also consistent with this account. Organizations with this structure of support are perceived to be highly credible members of their market cores. Such organizations, we predict, will have the leeway to react most strongly when their status falls. The nature of their connections gives them an implicit license in their price-setting decisions that other peripheral organizations may lack. Thus we expect:

Hypothesis 2: A focal organization increases price more substantially in reaction to status loss when that organization appeals broadly to disparate kinds of buyers.

We also expect the positive effect of status loss on its future price to be greatest for the focal producer whose structurally equivalent rivals (Burt, 1987;

White, 2002) already charge relatively high prices. A focal producer is structurally equivalent to another producer if it has similar relationships with the same third parties, such as suppliers, buyers, or other constituents (Burt, 1992; cf. Lorrain and White, 1971; McPherson, 1983). When joined with behavioral research on fairness norms in markets, network-analytic research on structural equivalence provides clear guidance on how a focal producer's closest rivals amplify (or dampen) reactions to slips in status.

An important implication of behavioral research is that rivals' past prices shape the normative context in which a focal producer selects its own price. Kahneman, Knetsch, and Thaler (1986) observed that whether buyers perceive a given producer's price as unfair hinges on the prices already set by that producer's rivals. Rivals that have previously charged a high price afford the producer greater latitude in subsequent price setting, allowing that producer to charge more without transgressing norms of fairness. Status-aspirational pricing can become a viable strategy when the focal producer is unconcerned about exceeding its peer-based ''just price'' (Thaler, 1985: 205), which could lead to being stigmatized as unfair or even exploitative (Bolton, Warlop, and Alba, 2003). Support for this claim also appears in the marketing literature around the concept of ''contextual reference prices'' (Rajendran and Tellis, 1994) and in sociological research on markets. In particular, White's (2002) formal model posits that all producers in a given market offer buyers an equally fair deal and suggests that penalties await the producer that exceeds the price expected in light of its peers' pricing decisions.

One particularly valuable and precise approach to identifying such peers relies on the concept of structural equivalence. Work on the subject also supports the contention that structurally equivalent peers shape a focal producer's autonomy to react to a status decline by raising its price. Earlier work (Merton, 1957; Burt, 1982) proposed structural equivalence as the social architecture out of which reference groups emerge in competitive domains, while also demonstrating that similarly situated rivals define appropriate conduct for a given organization (Mizruchi, 1990; Galaskiewicz and Burt, 1991; Bothner, 2003). Research in this vein carries direct implications for the circumstances in which a focal producer reacts to status loss by increasing its price. Conditioning on a given organization's prior price, we expect an organization to react most assertively to status decline when its nearest peers have previously charged high prices. Under this scenario, the organization has the freedom to increase price with a lower risk of violating buyers' normative expectations. We predict:

Hypothesis 3: A focal organization increases price more substantially in reaction to status loss when that organization's structurally equivalent rivals have previously charged high prices.

METHOD

Context and Variables

We tested our hypotheses using a panel of nationally focused liberal arts colleges and universities covered by *U.S. News & World Report* (hereafter *USN*) from 2005 through 2012. This empirical context is appropriate for assessing our predictions for three main reasons. First, it offers a clear measure of organizational status—the *USN* rank—and the opportunity to examine its effects on

organizations' pricing decisions. Second, it contains organizations for which status is a goal and whose status positions change meaningfully over time. Specifically, *USN*'s separation of liberal arts colleges and universities by geographic scope and degree offerings, based on the Carnegie Foundation's classification system (http://classifications.carnegiefoundation.org), allowed us to focus on schools for which status matters most: institutions with an (inter)national reach. Our panel excludes institutions covered by non-national *USN* rankings, such as regional and baccalaureate schools, because although status and tuition are important for these schools, they tend to have a narrower strategic purview, monitoring peer schools and targeting students within a close geographic radius. Third, during the seven-year interval from 2005 through 2011, *USN* reported detailed interorganizational relational data based on reports of incoming students' application behavior, which allowed us to construct the network-related covariates required by our two moderating hypotheses.

***USN* rankings.** *USN* creates a hierarchy of ranks each year based on weighted and measurable (if contested) inputs. The ranks are a public, manufactured reflection of the prestige of colleges and universities (hereafter "schools") as viewed by administrators as well as statisticians at *USN*. Owing both to their involvement in the rankings' calculation and to the fact that their jobs often hinge on improving their schools' positions in this ordering (Farrell and Van Der Werf, 2007), administrators, provosts, college presidents, and board members are thought to care more about the rankings than does any other constituency (Gordon, 2012). These individuals are also responsible for setting tuition levels, though the specific office or position assigned to the task differs by school.

USN began ranking undergraduate institutions in 1983, basing those rankings exclusively on the subjective assessments of college administrators. Today, *USN* rankings reflect a much more intricate process, though they are no less a subjective "gauge of status" (Nocera, 2012: 2), as the magazine still depends disproportionately on the opinions of top administrators when calculating ranks. *USN* takes the weighted average of a series of scores, ranging from subjective appraisals and student selectivity to alumni giving, to create an "overall score" for each school, which is then normalized to 100 for the top-ranked college and top-ranked university. The normalized scores are converted to ordinal positions and published as the final annual *USN* rankings.[1]

In 2005, Williams, Amherst, and Swarthmore received overall scores of 100, 98, and 98, respectively, and accordingly occupied ranks 1, 2, and 2 in the college category. In that same year, for universities, the overall scores of Harvard,

[1] For example, *USN* employed the following seven-component weighting system when calculating its 2005 rankings of the colleges and universities in our panel. First, 25 percent was applied to the results of an "academic peer assessment" questionnaire that asked administrators from each school surveyed to score all other schools with which they were familiar; second, a 20 percent weight was attached to the student retention rate; and third, 20 percent was applied to an aggregation of variables under the heading "faculty resources," which includes class size and faculty salaries. The remaining four components received the following weights: 15 percent to an aggregation of "student selectivity" measures (proxies for student quality, such as SAT scores), 10 percent for the school's financial resources, 5 percent for its graduation rate, and 5 percent for the giving levels of its alumni. The weighting schemes changed slightly over the course of our panel, but the comprising categories did not.

Princeton, and Yale were 100, 100, and 99, and so they held ranks 1, 1, and 3. Ties—like the one in 2005 between Harvard and Princeton—are even more common as ranks grow larger (e.g., seven universities were tied for 74th in 2005). In addition, ties occur by construction beyond the first tier. As an example, *USN*'s 2005 *Best Colleges* assigned ranks for colleges ranging up to 105 (where there was a five-school tie). Without explicitly denoting a second tier, the guide noted that schools in the third tier were ranked between 111 and 161 and were listed alphabetically instead of by rank. Fourth-tier colleges were also listed alphabetically and were listed as being ranked from 163 to 215. Given *USN*'s approach, we followed Sauder and Lancaster's (2006) method for assigning ranks to schools outside of the top tier: any school listed alphabetically by group was assigned the average rank (rounded to the nearest integer) of the reported range for that group. We also created a corresponding dummy variable, *not-first-tier*, for any school with a rank assigned by this method.

Yearly contests for status among schools. Our empirical setting is also marked by a dual condition that must be met for our hypotheses to have relevance in a given context: status shifts across time, and status positions are important objectives for the organizations involved. This condition characterizes the nationally focused schools covered by *USN*: college administrators are perhaps extraordinary in their fascination with prestige and their obsession with yearly changes in standing. Although financial health is certainly a concern for administrators and overseers, many colleges and universities are arguably more concerned with maintaining and raising their status than anything else (James, 1990). Unsurprisingly, in the first formal economic model of the modern university, Garvin (1980) depicted academic departments (and by extension, entire schools) fundamentally as status maximizers (see also Ehrenberg, 1999). More so than in other industries in which status also matters—such as banking (Podolny, 1993; Jensen, 2003)—in higher education, maximizing institutional status is a primary aim: ''if colleges and universities have a single-valued objective function, it is something like 'prestige maximization'" (Winston, 1999: 16). Academics' reflections on their own context are paralleled in more popular accounts. Caught up in the *USN* ''rankings craze'' (Lovett, 2005), administrators are thought to ''obsess over every incremental shift on the rankings scoreboard'' (Pope, 2012), while college presidents ''live and die'' by *USN* rankings (Stecklow, 1995).

There is, though, an important caveat here: public institutions often have different missions from those of their private counterparts. In particular, public schools are often more committed to the objective of equal access than the goal of relative standing. Consequently, in the empirical models we present, our estimates reflect the rankings and tuition levels of only private institutions.[2]

[2] There is a second compelling reason for splitting our panel by private versus public school: forces inside and outside public institutions, including state legislatures, often directly shape their tuition-setting behavior (Ehrenberg, 2002). Although various groups set tuition at private schools—at times the board of trustees or budget office, in other cases the provost or a tuition committee—tuition is always set inside the institution and apart from state-level political intervention (Brewer, Gates, and Goldman, 2001). Unlike their private counterparts, which can adjust tuition flexibly in reaction to market stimuli, public schools are often constrained by state-specific tuition plans that extend over several years. Therefore, though public institutions are incorporated in the inter-school network covariates to which we now turn, they do not appear in related panel models.

Status loss and status gain. Using the *USN* rank as our starting point, we constructed two dummy variables, *status loss* and *status gain*. Estimating the effect of a loss in status on future tuition is necessary for testing our three hypotheses. Conditioning separately on a second indicator that records a significant gain in status is important for addressing alternative explanations for the empirical pattern we observed, which we describe below.

Our measurement strategy follows earlier work in performance feedback theory in which researchers have relied on important facets of an organization's history (Greve, 1998) and have focused on deviations from an aspiration level or reference point (Baum et al., 2005). In particular, we build on Baum and colleagues' (2005: 538) conception of an organization's reference point as a "boundary between perceived success and failure." Using this imagery as our foundation, we assumed first that the focal school i is relatively neutral or indifferent when its rank in year t, R_{it}, is reasonably proximate to its school-specific, time-invariant mean rank, $\overline{R_i}$. Unlike very long (i.e., multi-decade) panels in which several time-varying reference points likely arise for each organization, our panel contains seven years, 2005–2011, along which we can measure our explanatory variables. Consequently, in our setting, a focal school's average rank is a plausible midpoint around which to proxy the width of that school's time-constant and latent boundary between perceived failure and success in the rankings. We set the width of this boundary at ten ranks. When a given school is inside a ten-rank range around its mean, it occupies a region of relative indifference, compared with the sharp dissatisfaction (or satisfaction) it would sense were it more than five ranks worse (or better) than this mean. We used a ten-rank range so that yearly changes in status captured by our dummy variables exceed white-noise fluctuations, while also not being so broad that virtually no schools experience these changes during our observation window. More generally, we viewed the organizations in our setting as led by relatively boundedly rational decision makers who do not pinpoint a single, precise rank at which they are indifferent. They instead know when they reside meaningfully outside (below or above) a comfortable range in the tournament for ranks.

Our measures of status loss and status gain are each simple binary switches that reflect two events. Status loss equals 1 rather than 0 under these two conditions: first, school i is currently (in year t) more than five ranks worse than its school-specific mean; second, school i previously (in a year $p < t$) ascended above that mean. Our measurement of status gain follows the same logic, switching from 0 to 1 when these two mirror-image conditions are met: school i is currently (in year t) more than five ranks better than its school-specific mean, and school i historically (in a year $p < t$) fell beneath that mean. We depict these two measures as follows:

$$statusloss_{it} = 1 \text{ if } R_{it} > \overline{R_i} + 5 \cap R_{ip} < \overline{R_i},\ p < t;\ 0 \text{ otherwise} \tag{1}$$

$$statusgain_{it} = 1 \text{ if } R_{it} < \overline{R_i} - 5 \cap R_{ip} > \overline{R_i},\ p < t;\ 0 \text{ otherwise} \tag{2}$$

Conditioning on status gain, together with status loss, is important for two reasons. First, we are interested in testing the prediction that a status decline triggers a higher future price—an effect that could instead be traceable to an increase in status that influences decision makers to lower price or hold it constant. Such an outcome might occur if a status gain ratifies existing behaviors,

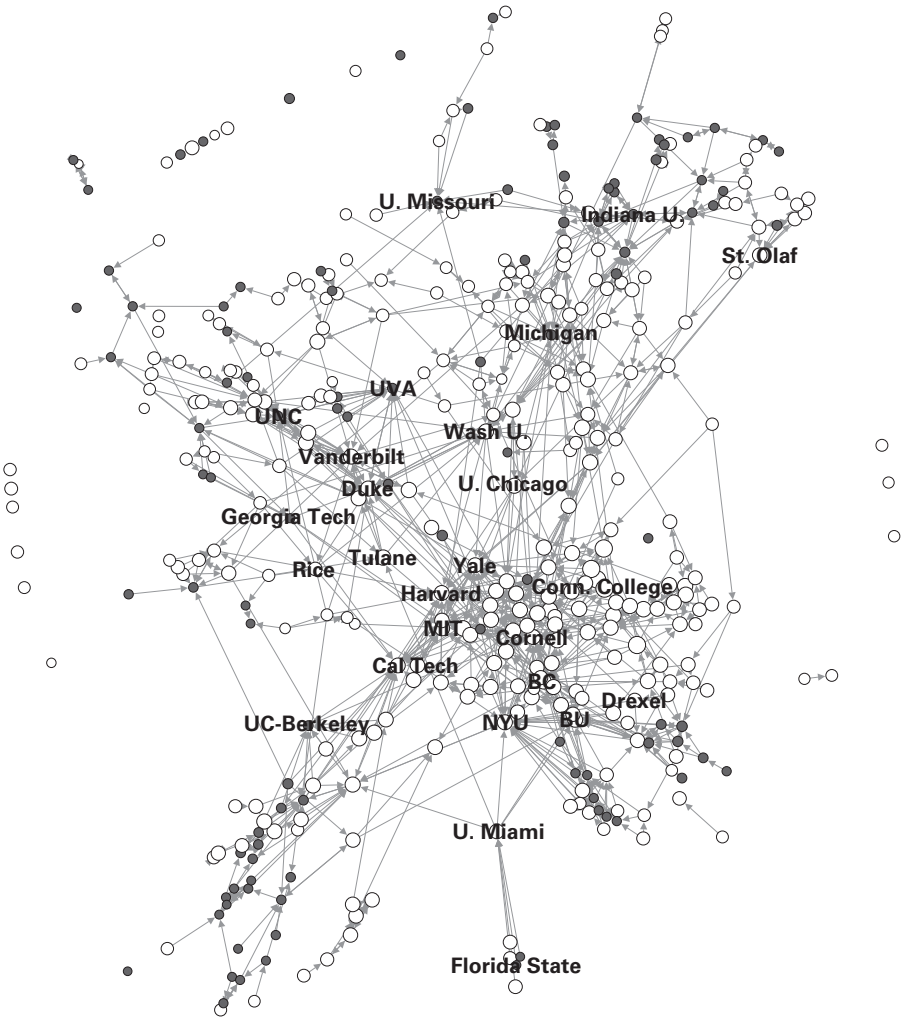
fostering complacency (Bothner, Kim, and Smith, 2012) and solidifying the sense that change is unnecessary (Audia, Locke, and Smith, 2000). Second, turning to specific features of our empirical setting, an improvement in status might also reduce the chances of raising tuition because higher status brings greater visibility (Sauder, Lynn, and Podolny, 2012) and thus stronger normative pressure to seem less expensive. In a setting in which political leaders have felt inspired to ''declare . . . war on the rising cost of college'' (Weissmann, 2012), it is important to consider that increasing status may make a focal school more susceptible to an ''outrage constraint'' (cf. Bebchuk, Fried, and Walker, 2002) and therefore averse to raising tuition. Entering status gain as an adjustment allows us to address these possibilities and, in particular, to disentangle the effects of loss versus gain.

Moderating variables: Betweenness centrality and weighted peer tuition. Inter-school relational data reported by *USN* allowed us to capture the width and diversity of schools' positions in the college-applicant market. Our measure of betweenness centrality (Freeman, 1979; Brandes, 2001) based on these data permitted us to identify credible core institutions to test hypothesis 2. Summarizing our calculation of betweenness centrality, we first note that from 2005 through 2011, *USN* listed the top five other schools to which each focal school's students also applied. For example, in 2009, New York University (NYU) cited these five schools as those to which its students also applied: Boston University, Columbia University, Cornell University, University of Southern California, and the University of Pennsylvania. Columbia University in turn cited these five institutions in 2009: Harvard University, MIT, Princeton University, Stanford University, and Yale University. Using these citations as inter-school ties, we pooled all private and public schools in each year and generated symmetric, annual inter-school citation matrices $\mathbf{A}_t$, where $\mathbf{A}_t = [a_{ijt}]$. In this matrix, $a_{iit} = 0$, $a_{ijt} = 1$ if school i cited school j or school j cited school i as an overlap in year t, and $a_{ijt} = 2$ if i and j jointly cited each other.

We then applied Brandes' (2001) method to $\mathbf{A}_t$ to generate betweenness centrality scores for all schools in each year. The betweenness centrality of a given node in a network is a function of the number of shortest paths between two other nodes passing through the focal node. In our setting, a school with a high level of betweenness centrality—such as Boston University (BU), NYU, or Washington University in St. Louis (WashU) shown in figure 1—reaches across diverse college applicants. Such schools tend to cite or be cited by schools that typically do not cite each other as overlap schools, meaning that they appeal broadly to a set of students whose application-related preferences are divergent. Unlike peripheral schools with limited market reach, occupants of these bridging positions in the network are more likely to be capable of aspirational pricing. Restating hypothesis 2 in our empirical setting, we expected the positive effect of status loss on future tuition to increase with betweenness centrality.

The inter-school relational data also allowed us to construct a measure of close rivals' past pricing decisions, which we term *weighted peer tuition*. This variable is important both as a conditioning variable and as a moderator for testing hypothesis 3. First, as a conditioning variable, *weighted peer tuition* allows us to account for a dynamic feature of status-aspirational pricing: a focal school

Figure 1. The 2009 college and university applicant network.*



* Clear nodes are private schools; dark nodes are public schools. Node size is based on tuition.

could, in principle, raise its price not only in reaction to a sharp drop in rank but also as a reaction to seeing its closest rivals pricing high. Not accounting for rivals' past prices could consequently lead us to overestimate the effect of a status decline, especially if a focal school fears that rivals' higher prices are status-aspirational and will subsequently hurt the focal school's rank. Controlling for these rivals' past pricing behavior thus helped us disentangle the effect of a status loss from rival-driven fears of future status descent.

Second, as a moderator, *weighted peer tuition* permitted us to test our third prediction, that the positive effect of status loss on future tuition rises with the lagged tuition levels of these peers. Our approach to measuring this follows Burt (1987, 1988). We first calculated two Euclidean distances in citation patterns between the focal school and all other schools tracked by *USN* in the

focal year. We computed these distances from yearly-varying matrices of inter-school citations $\mathbf{C}_t$ where $\mathbf{C}_t = [c_{ijt}]$, $c_{iit} = 0$, and $c_{ijt} = 1$ if school j cited school i as an overlap in year t.[3] Therefore, insofar as i and j have similar row vectors in $\mathbf{C}_t$, the same schools identify them as overlaps, and to the extent that they have similar column vectors, they identify the same schools as overlaps. Letting $dist_{ijt}$ denote the Euclidean distance between i's and j's rows plus the distance between i's and j's columns, we then computed the citation-based proximity of i and j as $prox_{ijt} = \max(dist_{ijt}) - dist_{ijt}$. After normalizing $prox_{ijt}$ for each school i, so that $w_{ijt} = prox_{ijt} / \sum_j prox_{ijt}$, and collecting tuition values for j, our measure of peer tuition may be written as:

$$weightedpeertuition_{it} = \sum_j w_{ijt} tuition_{jt} \tag{3}$$

Estimation and Conditioning Variables

We estimated the effect of status decline on schools' future tuition using two-way fixed-effects models (e.g., Podolny, Stuart, and Hannan, 1996) of the form:

$$log(Y_{i,t+1}) = \rho log(Y_{it}) + X_{it}\beta + \theta_1 statusloss_{it} + \theta_2 statusgain_{it} + \sigma_i + \tau_{t+1} + e_{i,t+1} \tag{4}$$

where $Y_{i,t+1}$ denotes school i's tuition in year $t+1$ and t ranges from 2005 to 2011. We included lagged tuition for three reasons. First, owing to inflation, growing demand for campus services and amenities, and the perceived need to price high for the sake of prestige, schools rarely hold tuition levels constant. Schools typically raise tuition every year; the question is by how much. Second, tuition is remarkably state-dependent (''price stickiness'' is substantial), and we therefore wished to estimate the effect of status decline conditional on the prior level of tuition. Third, a dynamic model isolates the unique consequences of a slip in rank, separate from a school's proximity to an overall price ceiling (captured jointly by lagged tuition and the year fixed effects).[4]

[3] The matrix $\mathbf{C}_t$ used to calculate weighted peer tuition differs from $\mathbf{A}_t$ used to compute betweenness centrality. More precisely, $\mathbf{A}_t = \mathbf{C}_t + \mathbf{C}_t^T$. Using a symmetric matrix of 0's, 1's, and 2's meant that more information entered our calculation of betweenness centrality. Using the rows as well as the columns of $\mathbf{C}_t$ similarly allowed us to incorporate more information into our structural-equivalence-based measure of peers' past tuition levels than using incoming or outgoing citations alone: the rows of $\mathbf{C}_t$ reflect other schools' citations of the focal school, while the columns of $\mathbf{C}_t$ reflect the focal school's citations of others. The columns may, in part, capture the aspirations of the focal school (a school may ''cite up,'' which is relevant for capturing which alters it eyes when setting its own tuition), while the rows, in aggregate, represent the (possibly more objective) information on position emanating from other schools.

[4] Because we estimated models of yearly tuition growth, we took two important steps. First, we excluded from our models any tuition values for which *USN* published an old or imputed tuition. When a school does not report a current tuition level to *USN*'s annual survey, the publication prints the previous year's tuition and indicates that it has done so. This occurs for 7.1 percent of all private school tuition values and for 4.2 percent of first-tier private school tuition values. Second, we estimated several robustness check models that incorporate corrections for first-order autocorrelation, using Stata's *xtregar* command. Our estimates of interest remained significant at the conventional level of confidence.

We also included several adjustments in the matrix $\mathbf{X}_{it}$ whose coefficients are in β. Table 1 includes correlations and descriptive statistics for the private schools in our panel models. We lagged all conditioning variables by one year so that they correspond to the measurement year of our main covariate of interest, status loss.[5] Summarizing our adjustments in $\mathbf{X}_{it}$, we first entered the indicator variable *not-first-tier* to denote those schools not included in *USN*'s top tier and for which we assigned the average of the upper and lower ranks that mark its tier.

Our second set of controls is related to student-body demographics, starting with size. We adjusted for student-body size because if a school were to decline in status and size simultaneously, that school might then raise tuition not just to improve in standing but also to cover the related loss in forecasted revenue. In addition, we captured facets of the demographic composition of the student body in two ways: through percentages of ethnic minorities and by Simpson's diversity index. Controlling for demographic diversity is essential because diversity may be associated with status (Stevens, 2007) while also tempering the rate of tuition growth—potentially reflecting the consequences of an enrollment-management strategy geared toward greater inclusiveness (DesJardins, Ahlburg, and McCall, 2006; Bowman and Bastedo, 2009). We collected these percentages of minorities from the government-run Integrated Postsecondary Education Data System (IPEDS). Categories omitted from tables 1 and 2 are percentages for multiethnic, white, and unknown. We computed Simpson's diversity index as the inverse of the sum of the squared proportions of all eight ethnic categories tracked by IPEDS. We expected this index to affect future tuition negatively.

We also conditioned on two indicator variables related to inter-school citation data. *Zero in-degree* was included to ensure that the effect of betweenness centrality is not a consequence of not being cited by other schools as an overlap. *Zero out-degree* addressed the possibility that when a school chooses not to send network data to *USN*, it pays less attention to its rivals and sets prices differently. When a school does not respond to this aspect of *USN*'s survey, this could reflect less susceptibility to the pricing decisions of other schools, and its own rate of tuition growth may change as a result.

A fourth set of controls was included to adjust for consumer demand. We entered the average SAT scores of incoming students because willingness to pay may rise with the opportunity to associate with other high-scoring students (see Hansmann, 1999, on education as an "associative good"). Schools whose current students have high SAT scores should be more attractive to prospective students. Consumer demand for a particular school may also be proxied by

[5] We chose a one-year lag structure in light of the way that *USN* collects and reports data. To provide a concrete example, one of the seven academic years in our panel for which we predict tuition is the 2007–2008 academic year; 2008 is thus year $t+1$ in equation (4). The focal school's level of tuition for the 2007–2008 academic year is published by *USN* in August 2007, based on the tuition level set by that school typically during the 2006–2007 academic year and often announced in letters to parents in the spring of 2007. Given a one-year lag, this level of tuition is predicted as a function of the (loss in) rank published by *USN* in the prior guidebook, which was published in August 2006. Our models assume that the private colleges and universities in our panel are able to respond by the spring of 2007 to a status decline evident by the late summer of 2006. Although the multiyear tuition-setting plans of many public institutions do not lend themselves to this lag structure, private institutions' budgetary cycles and tuition-setting schedules are shorter and more flexible, making this an appropriate interval.

Table 1. Correlations and Descriptive Statistics for Variables in Analyses

Variable	Mean	S.D.	Min.	Max.	1	2	3	4	5	6	7	8
1. Log (tuition)	10.18	0.31	8.10	10.81								
2. Rank	97.95	65.13	1	230	–.62							
3. Status loss	0.13	0.34	0	1	–.15	.43						
4. Status gain	0.06	0.23	0	1	–.08	.07	–.12					
5. Not-first-tier	0.37	0.48	0	1	–.53	.85	.40	–.06				
6. Full-time students	2892	3166	67	27986	.01	–.26	–.11	–.07	–.25			
7. % American Indian	0.00	0.01	0	0.05	–.02	–.05	–.02	–.03	–.04	–.01		
8. % Asian American	0.06	0.06	0	0.4	.41	–.52	–.22	–.08	–.36	.32	.13	
9. % African American	0.08	0.16	0	1	–.46	.30	.08	–.04	.28	–.06	–.10	–.14
10. % Hispanic	0.05	0.05	0	0.38	.17	–.06	–.08	–.08	–.01	.19	.15	.42
11. % international	0.04	0.04	0	0.31	.18	–.21	–.03	–.02	–.10	.11	.07	.36
12. Simpson diversity measure	1.97	0.80	1	5.03	.30	–.24	–.11	–.10	–.10	.25	.12	.74
13. Average SAT	1198	143	610	1525	.62	–.90	–.33	–.10	–.71	.28	.10	.60
14. Log (avg. loan)	8.53	0.36	7.37	9.80	.04	.27	.16	.10	.15	.14	–.06	–.14
15. Log (avg. grant aid)	9.47	0.47	6.78	10.48	.83	–.74	–.18	–.09	–.56	.06	.06	.47
16. Endowment ($ 000s)	792401.8	2645291	0	36600000	.21	–.36	–.12	–.07	–.21	.26	.16	.40
17. Endowment missing	0.19	0.39	0	1	–.47	.49	.22	.01	.42	–.15	.01	–.26
18. State GDP ($ 000,000s)	512458.8	438505.8	20537	1900463	.19	–.11	–.04	–.01	–.08	.09	.08	.40
19. Betweenness centrality	1847.51	3217.884	0	35468.23	.18	–.26	–.10	–.05	–.17	.49	–.03	.29
20. Zero out-degree	0.12	0.33	0	1	–.15	.19	.07	–.02	.22	.02	–.04	.00
21. Zero in-degree	0.31	0.46	0	1	–.38	.54	.17	.04	.51	–.27	.05	–.17
22. Weighted peer tuition	18521.4	1935.113	15236.61	21742.63	.22	.09	.22	.14	.03	–.02	.06	–.01
23. Middle top school	0.23	0.42	0	1	.04	–.18	–.16	.16	–.40	.12	.01	–.10
24. Rankpeeraid$_{t-1}$	10839.41	2582.594	4508.396	16591.84	.38	–.49	–.07	–.02	–.41	–.36	.00	–.04

Variable	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
10. % Hispanic	–.03														
11. % international	–.04	.13													
12. Simpson diversity measure	–.02	.66	.54												
13. Average SAT	–.42	.09	.28	.32											
14. Log (avg. loan)	.07	.00	–.06	–.06	–.28										
15. Log (avg. grant aid)	–.35	.10	.27	.33	.73	–.11									
16. Endowment ($ 000s)	–.02	.14	.22	.36	.43	–.10	.33								
17. Endowment missing	.37	–.08	–.03	–.11	–.44	.01	–.42	–.14							
18. State GDP ($ 000,000s)	–.08	.52	.05	.37	.11	.00	.10	.03	–.06						
19. Betweenness centrality	.03	.10	.11	.22	.29	.08	.20	.39	–.10	.05					
20. Zero out-degree	.17	.02	.07	.07	–.14	.00	–.16	.14	.18	–.05	–.02				
21. Zero in-degree	.19	.07	.01	.06	–.50	.04	–.37	–.17	.33	.05	–.19	.09			
22. Weighted peer tuition	–.02	.05	.01	.05	–.05	.42	.18	.00	.01	.04	.04	–.09	–.01		
23. Middle top school	–.10	–.05	–.03	–.13	.05	.11	–.02	–.12	–.17	–.04	–.03	–.15	–.19	.01	
24. Rankpeeraid$_{t-1}$	–.19	–.20	–.06	–.19	.38	–.21	.45	–.03	–.16	–.06	–.07	–.19	–.30	.30	.06

students' willingness to take on loans to attend. Using data collected from IPEDS, we included the (logged) per-student average loan and expected loan levels to be positively associated with future tuition.

Our fifth set of covariates was intended to gauge the financial well-being of each school. Entering financial measures ensures that we do not overestimate the unique effect of status loss: financial shortfalls may arrive jointly with slips in rank, resulting in tuition hikes as schools look to increase cash flow to

support their strategic objectives (Corkery, 2012). First, we entered (logged) *average institutional grant aid*, which includes all scholarships and fellowships granted to undergraduates by the institution itself, meaning the school uses this money to attract students it desires without expecting financial remuneration (i.e., this covariate excludes loans). Second (and third), we entered each school's endowment, collected from the National Association of College and University Business Officers (NACUBO), plus a dummy variable, *endowment missing*, indicating that the focal school had no record of an endowment according to NACUBO. Such schools' endowments were coded as zero. Controlling for endowment is important because schools with higher endowments have more freedom to adjust tuition levels, undertake projects, improve facilities, hire faculty and staff, and offer aid, all of which might directly improve rank and lower future tuition growth. We adjusted as well for the GDP of the state in which the school is located: more-favorable local economic conditions might associate a better rank with lower subsequent tuition because of increased revenues from other sources, such as student athletic events.

School fixed effects and year fixed effects, represented by σ_i and τ_{t+1}, are the final two terms in equation (4). Entering school fixed effects, so our estimates reflect within- rather than between-school variations, is essential for testing our main hypothesis that a given producer sets price higher in reaction to the perceived problem of status loss, for two reasons. First, the fixed effects sweep out unobserved stable traits such as preexisting reputation, location, or an unusually large athletics budget that might create the appearance of a positive effect of status on price. Second, they allow us to bring into focus the adaptive responses of a focal organization as it travels through its status hierarchy over time. Year fixed effects are included in all models because tuition rises on average each year. Year fixed effects also adjust for annual changes in the applicant pool, the number of schools to which the average student applies, and the health of the stock market, which dramatically affected college endowments over the course of our panel.

RESULTS

We present estimates for six models predicting future tuition for private institutions in table 2, which starts with a simple pooled cross-sectional model. We included only *USN* rank, our not-first-tier dummy variable, and year fixed effects. School fixed effects are omitted. In model 1, the coefficient on *USN* rank is negative, as schools positioned in the top ranks charge higher tuition. This result is consistent with previous work on status showing that higher-status organizations command higher prices for their goods and services.

In model 2, we estimated a within-school growth model by adding lagged log (tuition) and school fixed effects. Although we do not report coefficients on the school indicators to conserve space, these (available by request) express sharp between-school differences in pricing tendencies, confirming the merits of the within-school estimator. In addition, the coefficient on *USN* rank now is significant and positive. When our estimates reflect within-organization

Table 2. Estimates for Pooled Cross-sectional and Fixed-effects Models Predicting Log (Tuition), 2006–2012*

Variable	1	2	3	4	5	6
Log (tuition)		.564•••	.563•••	.538•••	.534•••	.536•••
		(.0794)	(.0784)	(.0683)	(.0681)	(.0680)
Status loss			.00925•••	.0102•••	.00610••	–.0418
			(.00243)	(.00258)	(.00266)	(.0260)
Status gain			.00137	.000147	.000152	.000784
			(.00269)	(.00289)	(.00289)	(.00292)
Rank	–.00325•••	.000169••				
	(.000134)	(8.11e-05)				
Not-first-tier	–.00615	–.00788•	–.00664	–.00789	–.00789	–.00668
	(.0221)	(.00422)	(.00489)	(.00548)	(.00546)	(.00575)
Full-time students				–1.79e-06	–1.28e-06	–1.81e-06
				(3.60e-06)	(3.64e-06)	(3.61e-06)
% American Indian				–.172	–.172	–.158
				(.173)	(.174)	(.173)
% Asian American				.109	.116	.115
				(.0807)	(.0808)	(.0802)
% African American				–.0710	–.0706	–.0748
				(.0612)	(.0624)	(.0609)
% Hispanic				.0799	.0700	.0778
				(.0720)	(.0725)	(.0723)
% International				.163••	.154••	.160••
				(.0758)	(.0770)	(.0759)
Simpson diversity measure				–.0126•••	–.0120••	–.0125•••
				(.00486)	(.00479)	(.00486)
Average SAT				.000165•••	.000167•••	.000169•••
				(6.14e-05)	(6.12e-05)	(6.17e-05)
Log (avg. loan)				.0108•••	.0108•••	.0106•••
				(.00288)	(.00289)	(.00286)
Log (avg. grant aid)				.0281•••	.0285•••	.0282•••
				(.0101)	(.0100)	(.0101)
Endowment ($ 000s)				–2.67e-09•••	–2.62e-09•••	–2.79e-09•••
				(7.97e-10)	(7.77e-10)	(7.93e-10)
Endowment missing				.00355	.00277	.00351
				(.00551)	(.00529)	(.00547)
State GDP ($ 000,000s)				1.31e-08	1.33e-08	1.42e-08
				(1.53e-08)	(1.53e-08)	(1.55e-08)
Zero out-degree				–.00177	–.00139	–.00100
				(.00435)	(.00437)	(.00439)
Zero in-degree				–.000811	–.000603	–.000680
				(.00285)	(.00283)	(.00286)
Weighted peer tuition				–2.43e-05•	–2.41e-05•	–2.15e-05•
				(1.31e-05)	(1.30e-05)	(1.30e-05)
Betweenness centrality				4.72e-07	1.96e-07	4.79e-07
				(3.32e-07)	(3.24e-07)	(3.31e-07)
Status loss × Betweenness centrality					3.68e-06••	
					(1.46e-06)	
Status loss × Weighted peer tuition						2.57e-06•
						(1.34e-06)
2007	.0661•••	.0303•••	.0303•••	.0554•••	.0556•••	.0529•••
	(.0215)	(.00590)	(.00580)	(.0151)	(.0150)	(.0150)

(continued)

Table 2. (continued)

Variable	1	2	3	4	5	6
2008	.131•••	.0523•••	.0520•••	.102•••	.102•••	.0964•••
	(.0218)	(.0103)	(.0101)	(.0321)	(.0318)	(.0317)
2009	.202•••	.0764•••	.0754•••	.149•••	.149•••	.141•••
	(.0202)	(.0148)	(.0144)	(.0468)	(.0465)	(.0463)
2010	.245•••	.0860•••	.0844•••	.191•••	.191•••	.178•••
	(.0201)	(.0193)	(.0188)	(.0676)	(.0671)	(.0669)
2011	.275•••	.108•••	.106•••	.234•••	.234•••	.218•••
	(.0236)	(.0228)	(.0223)	(.0827)	(.0822)	(.0818)
2012	.310•••	.124•••	.123•••	.269•••	.269•••	.251•••
	(.0265)	(.0259)	(.0254)	(.0939)	(.0932)	(.0928)
Constant	10.39•••	4.414•••	4.442•••	4.542•••	4.572•••	4.508•••
	(.0157)	(.796)	(.786)	(.754)	(.752)	(.750)
Observations	1,873	1,774	1,774	1,694	1,694	1,694
R^2	.454	.953	.953	.959	.96	.96

• $p < .10$; ••$p < .05$; •••$p < .01$.
* Robust standard errors are in parentheses. All covariates are lagged one year. Overall R^2 values are shown for model 1, within-R^2 values for within-school models 2–6.

variations only, a decline in status is associated with a higher future price. Using this very simple specification, we see initial support for our first hypothesis.

Model 3 replaces the *USN* rank with our measures of status loss and status gain. The coefficient on status loss is significant and positive, indicating that when a school slips more than five ranks beneath its time-constant, mean rank—after having ascended above that threshold previously—it then raises tuition in the next year. The effect is economically meaningful: the increase rounds to a full percentage point, which is the equivalent of the provost announcing in his or her spring letter to parents that next year's tuition will go up by 5 percent instead of by 4 percent.

In contrast, the effect of status gain is statistically indiscernible. Upward status mobility, which we thought might bind a school more tightly to normative constraints to charge students less, does not uniquely affect price. We instead see further evidence that downward status mobility is more influential in shaping pricing decisions. In addition, the asymmetric effects of status loss versus status gain imply that reversion to the mean is unlikely to account for the positive effect of status loss. If reversion to the mean were the main process at work, we would expect coefficients on these two variables that are roughly equal in absolute value, rather than the substantial difference we observe.[6]

[6] One can imagine a simple stochastic process in which the rank of the focal school varies yearly, from low to high, around its average rank—for example, due to unplanned variations in the student-to-faculty ratio. Imagine further that the school's rate of tuition growth wavers in the same way around its typical level—perhaps due to volatility in the composition of the tuition-setting committee. Random slips in rank could then be followed by randomly higher tuition increases the next year. If, however, this were the main process at work in our context, we would also expect to see gains in rank followed by significantly smaller tuition increases the following year. Because the effect of status gain is instead insignificant, we conclude that reversion to the mean does not mark the link between a loss in status and higher future tuition.

In model 4, we conditioned on a wide range of factors that are important for isolating the consequences of declines in status.[7] Status loss retains its significance in the presence of these covariates, including those for student-body diversity. Simpson's diversity index is negative and significant, reflecting a tempered rate of future price growth as a school commits to a more heterogeneous student pool. The sustained positive effect of a fall in status, controlling for student diversity, suggests that our main effect of interest is not a byproduct of a school growing less diverse and then deciding to exploit a locked-in student body with higher socioeconomic status. That status loss stays significant together with the size of the endowment is also noteworthy. Clearly, price-setting committees raise tuition in response to financial shortfalls, yet conditioning on this process, model 4 indicates that they still also react to sharp declines in rank.[8]

The third and final set of continuous variables in model 4 includes betweenness centrality and weighted peer tuition, the covariates we constructed to test the interaction effects stated in hypotheses 2 and 3. We turn to these in models 5 and 6, respectively, but note their main effects here. Aligned with our discussion of betweenness centrality as a measure of the degree to which a school is a credible market incumbent, its effect is positive, though not quite achieving significance. It suggests, inconclusively, that the more a school's reach cuts across disparate sections of the market for students, the more it charges. The effect of weighted peer tuition is not what we anticipated: instead of seeing evidence of contagious pricing, we interpret this coefficient as a possible reflection of efforts to differentiate.[9]

In models 5 and 6 we turn to two interaction effects, which clarify when status-aspirational pricing is most prevalent, as not all organizations in a market have a license to do so. Supporting hypothesis 2, that broad appeal across otherwise disparate buyers allows an organization to price aspirationally, the effect of status loss by betweenness centrality is positive and significant. Note also the weakened main effect of status loss (at zero betweenness centrality) in model 5. Though the effect of status loss is lower for a peripheral school, its effect grows in magnitude as betweenness centrality rises. For instance, after

[7] As a robustness check, we estimated an alternative version of model 4 using different measures of status loss and status gain based on a four-year moving-window average rank to proxy the midpoint of the focal school's region of indifference. We computed average rank from years t–4 to t–1, and then in year t set status loss (or status gain) equal to one, rather than zero, if the focal school was more than five ranks worse (or better) than this mean. Using these alternative measures, our results were substantively the same. In model 4, the coefficient on status loss was .005 with a t-test of 2.13. Status gain was again insignificant (0.28 t-test). The advantage of our current measures of status loss and status gain is that they assume that schools do not continuously update their reference points, in response to surprising environmental feedback, over a relatively narrow timeframe. One limitation of our approach, however, concerns out-of-sample forecasting as opposed to in-sample explanation. In particular, if our goal were to find the best measure of status loss for predicting unknown tuition in a future year, we could only compare root-mean-squared errors across models predicting tuition in the final year of our panel, not for all years.

[8] Our effects remain significant when we include lagged rank in our models, as well. Because lagged rank is consistently insignificant once we enter our conditioning variables, we do not include it in our reported models.

[9] Our calculation of weighted peer tuition includes tuition levels for peer schools that were reprinted by *USN* (in rare cases, see footnote 4) based on a prior year. These values for the focal school are necessarily coded as missing in our models. As an added check, we also recomputed weighted peer tuition without these imputed values, and our results were virtually identical.

a status loss, tuition rises by 1.3 percent when betweenness centrality is at its mean but by 2.5 percent for betweenness centrality at one standard deviation above its mean and by 3.7 percent at two standard deviations above its mean.

We also find support for our second moderating hypothesis in model 6. Our prediction, following from our assumption that norms of fairness constrain price-setting decisions, was that organizations whose close competitors have charged high prices have the latitude to pursue status-aspirational pricing. Showing support for hypothesis 3, the coefficient on status loss by weighted peer tuition is positive. The positive effect on tuition of a fall in status is strongest when a focal school's structurally equivalent peers have previously set tuition high. For instance, at one standard deviation above the mean of weighted peer tuition, a decline in status is associated with a 1.1 percent increase in future tuition, and for the maximum value of weighted peer tuition, a drop in status is associated with a 1.4 percent increase.

Robustness Checks and Extensions

We took four main steps to evaluate the robustness of our main finding that schools raise tuition after a status loss.[10] First, given our claim that only core, rather than peripheral, organizations can price aspirationally, we examined counts of status losses and gains together with average yearly tuition increases across a range of status categories: quartiles of lagged ranks and first-tier versus not-first-tier schools. If our interpretation of our main effect of interest is correct, there should both be a sufficient number of status losses among first-tier schools, and the impact of status loss will be concentrated there.

As table 3 shows, across our panel's private schools, there were 251 status losses (14.1 percent of the 1,774 school-years) and 140 status gains (7.9 percent). This asymmetry reflects *USN*'s record of demoting more than it promotes, as well as the expanding size of its coverage over time, usually by adding lower-ranked schools. This feature of *USN*'s design also reinforces the

Table 3. Status Losses, Gains, and Tuition Changes by Status Quartile and Tier

	Status subgroup	Status losses	Status gains	Avg. annual tuition increase
Rank quartile	Top (1–40)	1	0	5.0%
	Second (41–91)	24	39	5.5%
	Third (92–155)	79	80	5.9%
	Fourth (156–230)	147	21	5.0%
Tier	First tier	87	118	5.3%
	Not-first-tier	164	22	5.4%
	All private schools	251	140	5.4%

[10] In addition, to ensure that our effects of interest stayed statistically discernible with a correction for first-order error autocorrelation, we estimated alternative versions of models 4, 5, and 6 using Stata's *xtregar* command. In model 4, the coefficient on status loss was .01 (4.41 *t*-test). In model 5, coefficients and test statistics for status loss and for status loss × betweenness centrality were these, respectively: .007 (2.41 *t*-test) and 3.23e-06 (2.24 *t*-test). And in model 6, they were –.049 (–1.81 *t*-test) for status loss and 2.94e-06 (2.20 *t*-test) for status loss × weighted peer tuition.

importance of adding year fixed effects. Further, sharp status changes almost never occur in the top quartile of the rankings (just one school suffered a status loss, and no schools ranked 40 or better posted a status gain). This likely reflects *USN*'s effort to prevent non-credible churn at the peak of the rankings, as well as our conservative definitions of status loss and status gain.

Turning to differences between first-tier and not-first-tier schools, in the first tier, in which tuition rose by an annual average of 5.3 percent, there were 87 status losses and 118 status gains. For not-first-tier institutions, tuition increased yearly by 5.4 percent, and there were 164 status losses and only 22 gains. Status losses for both categories are consistent with our theory, but it is still important to determine where, in the distribution of ranks, the effect of status loss is most strongly concentrated. Model 7 of table 4 reports the results of our main model (model 4 in table 2) with not-first-tier schools removed. Although this effect is slightly dampened—there is no longer a full percentage point increase in tuition after a status loss—the coefficient on status loss stays significant (2.52 *t*-test). Among first-tier schools, which have clearly crossed a credibility threshold, we see continued evidence in support of our main hypothesis.

Our second set of robustness checks sought to guard against the possibility that a small handful of schools suffer status losses sequentially while also aggressively raising tuition to arrest a free-fall, thus biasing our estimates. Model 8 presents estimates for the subset of schools from model 7 with one difference—we excluded all schools experiencing more than one status loss—and again we see a robust effect of our main variable of interest. This result gives us added confidence that our results are not a reflection of a few outlying schools flailing as they slide down the rankings.

Third, we assessed our claim regarding the direction of the effect of status loss from several vantage points. One approach was to ask whether the effect of status loss is strongest in the middle of the first-tier rankings category. If a status decline triggers higher tuition, rather than vice versa, status loss should exert weaker effects in some regions of the ranks than in others, rather than affecting tuition uniformly. In model 9, which again includes only first-tier schools, we interacted status loss with a dummy variable termed *middle top school*. This variable equals one if the focal school's rank resides between 50 and a 20-rank distance from the lowest first-tier rank that year. The estimates in model 9 suggest that schools in the middle of the top tier are also most responsive to a sharp decline. Such schools are core members of the market, but they are also arguably the most anxious about status—neither entrenched incumbents of the top 50 nor newcomers to the first tier—and a sharp fall in rank appears to trouble them the most.

In addition to an instrumental variable analysis (see the Online Appendix, http://asq.sagepub.com/supplemental), we also explored whether tuition is associated empirically with future status loss: counter to our theory, a tuition hike could trigger a slip in rank if schools get punished for appearing greedy or unfair. Using a conditional fixed-effects logit on our full panel of private schools, we modeled status loss as a function of all explanatory variables from model 4. In this supplementary model, lagged log(tuition) is insignificant (.30 *z*-score test), further suggesting that reverse causality is probably not responsible for the effect of status loss on future tuition.

Table 4. Estimates for Robustness Check Models Predicting Log (Tuition), 2006–2012*

Variable	7	8	9
Log (tuition)	.751•••	0.679•••	.746•••
	(.0571)	(0.0482)	(.0558)
Status loss	.00817••	0.00993••	.00208
	(.00324)	(0.00465)	(.00405)
Status gain	–.00196	–0.00254	–.00221
	(.00316)	(0.00351)	(.00331)
Full-time students	–1.57e-06	–2.02e-06	–1.28e-06
	(2.88e-06)	(2.89e-06)	(2.95e-06)
% American Indian	.0624	–0.166	.0349
	(.218)	(0.214)	(.221)
% Asian American	.0777	0.0949	.0825
	(.0605)	(0.0586)	(.0598)
% African American	–.134••	–0.145••	–.137••
	(.0577)	(0.0602)	(.0591)
% Hispanic	.116	0.0926	.109
	(.0908)	(0.0882)	(.0901)
% International	.0835	0.0967	.0880
	(.0817)	(0.0840)	(.0824)
Simpson diversity measure	–.00162	–0.00332	–.00150
	(.00438)	(0.00358)	(.00441)
Average SAT	6.76e-05	6.96e-05	7.21e-05
	(6.89e-05)	(7.38e-05)	(6.97e-05)
Log (avg. loan)	.00251	0.000212	.00204
	(.00287)	(0.00310)	(.00288)
Log (avg. grant aid)	.0203••	0.0233•••	.0205••
	(.00912)	(0.00855)	(.00905)
Endowment ($ 000s)	–1.81e-09•••	–2.17e-09•••	–1.85e-09•••
	(6.36e-10)	(6.47e-10)	(6.31e-10)
Endowment missing	–.00186	–0.0113•	–.00111
	(.00519)	(0.00656)	(.00526)
State GDP ($ 000,000s)	3.36e-09	–3.36e-10	4.86e-09
	(1.27e-08)	(1.19e-08)	(1.28e-08)
Betweenness centrality	3.55e-07	3.08e-07	3.46e-07
	(2.91e-07)	(2.76e-07)	(2.91e-07)
Zero out-degree	–.00137	–0.00274	–.000863
	(.00450)	(0.00453)	(.00458)
Zero in-degree	–.00145	–0.00244	–.00158
	(.00282)	(0.00293)	(.00279)
Weighted peer tuition	–6.31e-06	–1.01e-05	–4.58e-06
	(1.09e-05)	(1.11e-05)	(1.11e-05)
Middle top school			.00112
			(.00269)
Status loss × Middle top school			.0109••
			(.00514)
Constant	2.386•••	3.181•••	2.400•••
	(.651)	(.556)	(.644)
Observations	1,170	1,019	1,170
R^2 (within)	.978	.978	.978

• $p < .10$; •• $p < .05$; ••• $p < .01$.

* Robust standard errors are in parentheses. School and year fixed effects are included. All covariates are lagged one year.

Fourth, we addressed the possibility that status decline prompts schools to spend exorbitantly and thus to raise tuition for reasons unrelated to signaling. As schools fall in the rankings, they may expect the imminent loss of higher-caliber applicants to further hurt their rankings, leading them to employ various recruiting strategies to entice their preferred students. Such strategies could include providing more grants, improving the student-to-faculty ratio, or pushing fundraising efforts—each of which necessitates spending more money, which would likely come from increased tuition. We explored each of these scenarios among the ranked schools in our data (results available by request) and found that when (logged) average grant aid, the student-to-faculty ratio, and the alumni giving rate were each inserted as dependent variables in alternative versions of model 4, status loss was not a significant predictor for any of them (*t*-tests for status loss, across these three models, were .26, 1.13, and .02, respectively). Though certainly not conclusive, it appears that raising tuition following a decline in status is not—at least in the short term—related mainly to spending more money to attract more desirable students.

Qualitative accounts. Lastly, we collected and coded qualitative accounts of the Chivas Regal strategy in higher education to assess from another angle our claim that schools increase their prices to send a status signal. Although there has been discussion since the late 1980s (e.g., Werth, 1988) of status-aspiring schools hiking tuition to enhance perceptions, a second version of the Chivas Regal strategy in our context refers to investment in status symbols, such as new dormitories and gyms, through the revenues provided by tuition increases. This is a long-run version of status-aspirational pricing in which conspicuous consumption serves as the expected link between charging more and eventual status gain. We cannot rule out the possibility that this long-run version is also at work in our panel, but qualitative evidence suggests that the short-run, signaling-based version of the Chivas Regal strategy we have described typically summarizes the aims of those setting tuition.

Ideally, we would have direct and accurate data on the deliberations of those responsible for setting tuition each year to further test our theory. Unfortunately, though some university presidents, such as Stephen Trachtenberg of George Washington University (Luzer, 2010; Carey, 2015) and Richard Freeland of Northeastern University (Kutner, 2014), have discussed their use of the Chivas Regal strategy, such open revelations have typically come only after they have left the job. The price-setters themselves are generally unreliable sources when it comes to publicly confirming the use of higher prices to signal status, as the implicit understanding is that to reveal the strategy as such would be to kill its efficacy or incite outrage at the school.

Journalists, however, are not bound by these constraints. To check our main assertion, we searched Google and Lexis Nexis to collect all of their electronically available journalistic articles explicitly mentioning "Chivas Regal" in connection with higher education. Searching from 1/1/1980 through 1/1/2015, we identified 40 articles and one congressional hearing document: 31 documents from Lexis Nexis, six from Google News, and four from Google Web Search. We counted the total number of words in each document referring explicitly or implicitly to the Chivas Regal strategy and coded each sentence according to how the Chivas Regal strategy was characterized. Our approach yielded four different characterizations: raising tuition to

signal higher status, raising tuition to invest in status symbols, charging what the market will bear, and other or non-pertinent. (Sentences coded as "other" typically either characterized the Chivas Regal strategy ambiguously or conflated multiple characterizations, while sentences coded as "non-pertinent" did not imply any particular characterization.) Of the total words across all 41 documents, 6 percent were relevant in that they pertained to the Chivas Regal strategy. Of these relevant words, 64 percent pertained to tuition increases geared toward signaling higher status, while only 5 percent pertained to tuition increases to invest in status symbols. Some examples of text aligned with our emphasis on status signaling via higher tuition include the following:

> Says [Martin] Meyerson [UPenn's president from 1970–1981]: "We were building up a kind of notion about colleges and universities that the higher the price, the better they were." (Larson, 2001)

> For decades, most private college pricing has reflected the Chivas Regal effect—the notion that whether in a Scotch or a school, a higher price indicates higher quality. "Schools wanted a high tuition on the assumption that families would say that if they're charging that high tuition, they must be right up there with the Ivies," said David L. Warren, president of the National Association of Independent Colleges and Universities. (Lewin, 2013)

> The increase not only would allow the system to recover more of the cost of a student's education, he said, but also would encourage applicants to view College Park as a more selective institution, a concept known in higher education circles as the 'Chivas Regal argument.' "The perception is, if tuition's too low, people say, 'Hey, that school can't be any good,'" Hug said. "The best way to improve your reputation is to increase tuition." (Montgomery, 2003)[11]

The results of our qualitative search thus indicate that the strategic raising of prices not only occurs among status-aspiring schools but also appears to be done more in the name of the price–status perception link than to be able to invest in tangible status symbols. It is a well-known and regularly utilized tactic for university presidents and other price-setters.

Exploring consequences. These analyses lead us to an important final question: does the Chivas Regal strategy actually work? To try to address this question, our approach was to see if tuition is positively associated with future improvements in *USN* rank after a status loss. Table 5 presents conditional fixed effects logit models predicting the probability of a focal school's improvement in rank from year t to $t+1$ as a function of its (logged) tuition in t, and whether it suffered a loss in status in t–1. We used this estimator instead of a standard logit with dummies for schools to avoid the incidental parameters problem (Greene, 2004). For models 10–12, the outcome variable equals 1 under these conditions, respectively: the school improved by one or more

[11] Calvert (2006), in contrast to these statements, offers a characterization consistent with the investment version of Chivas Regal: "Chester Finn, a former Department of Education official, once toured several campuses with his daughter and was greeted everywhere with the 'Chivas Regal strategy' (as one dean called it), which lures students with amenities such as posh apartments, food courts and fancy gyms. Finn's daughter said it was 'like comparing resort hotels.'"

Table 5. Estimates for Conditional Fixed-effects Logit Models Predicting Rank Improvements*

Dependent variable	(10) Improve ≥ 1 rank	(11) Improve ≥ 2 rank	(12) Improve ≥ 3 rank
Rank	.0816•••	.0756•••	.148•••
	(.00990)	(.00986)	(.0166)
Log (tuition)	–1.101	2.502	–3.472
	(2.499)	(2.909)	(3.788)
Status loss	–16.11•	–24.27•••	–2.904
	(8.663)	(9.173)	(10.44)
Status gain	–.0795	–.126	.391
	(.343)	(.349)	(.392)
Not-first-tier	–1.408•••	–.944•••	–2.689•••
	(.358)	(.354)	(.480)
Status loss × Log (tuition)	1.548•	2.324•••	.339
	(.848)	(.897)	(1.018)
Observations	1,315	1,169	986
Number of groups	259	222	188

• $p < .10$; •• $p < .05$; ••• $p < .01$.
* Standard errors are in parentheses. Status loss and status gain are lagged two years; other predictors are lagged one year. Year fixed effects are included in all models.

ranks, two or more ranks, and three or more ranks. In all three models, the coefficient on the interaction term is in the expected direction, but only in model 11 is prior tuition strongly associated with future rank improvement when the school has previously undergone a loss in status. In contrast, the interaction between lagged status loss and tuition falls shy of significance in model 12, which predicts the probability of a rank improvement of three or more. Hiking tuition in the wake of the status loss appears to be beneficial for modest recovery but is in no way a panacea for schools seeking instant recovery from a sharp status loss.

DISCUSSION

Our examination of colleges and universities in the United States reveals an important and counterintuitive finding: an institution's loss of status will, under certain conditions, lead to status-aspirational—rather than status-consistent—pricing. Having clarified two conditions in which organizations raise prices in reaction to drops in status, we show that the impact of status on price can no longer be seen as exclusively positive. This revelation has implications for status theory, performance feedback theory, and administrative research on ranking systems.

We see two interlinked implications for status theory. First, new research will benefit from focusing more explicitly on the consequences of significant status changes. Considering both status declines and gains (see Azoulay, Stuart, and Wang, 2013; Jensen and Kim, 2015) as central explanatory variables in future work will broaden our understanding of how status shapes organizational conduct and performance. Such an empirical strategy may also require a broader portrait of the nature of status itself. Some of the most compelling studies, most notably Podolny's (1993) original formulation, have construed status as a stable asset yielding concrete benefits.

We can contrast this perspective with our own using two of Merton's (1968) metaphors. On one hand, similar to Podolny (1993), Merton pictured high-status actors as owners of an almost irrevocable "coin of recognition," an image characterizing status as a resource (cf. Saloner, Shepard, and Podolny, 2001: 53–55) that passively brings advantage. On the other hand, Merton also analogized a status position to a "way station" en route to the next "summit," which suggests that status is a rank in a tournament in which advances and reversals occur; status, here, is less like a resource and more like prized territory in a turf war. This dynamic imagery is aligned with more recent frameworks (e.g., Sauder and Lancaster, 2006; Espeland and Sauder, 2007) in which status shifts meaningfully in real time, confirms and threatens actors' identities (Marr and Thau, 2014), and requires constant cultivation. Taking an explicitly dynamic, within-organization view and zeroing in on a focal firm's reactions to sharp declines will likely cast new light on how organizational decision making varies with relative standing in ways that pooled cross-sectional designs cannot.

The second implication for status theory is the vital need to reconsider the link between status and agency. Prior work has underscored elites' latitude to choose non-normative actions (e.g., Han, 1994; Phillips and Zuckerman, 2001), observing that there are certain activities in which only high-status actors are allowed to engage. Although we agree with this in general, our results also suggest that agentic behavior is especially likely among those that have fallen in status. For instance, although Stuart (1998) found that increases in status raised the propensity to collaborate across firms, we suspect that a focal firm that was recently pushed down in status beneath its reference point may be driven, perhaps frantically, to form alliances. Using measures and methods similar to what we have proposed, it may be possible to demonstrate that the strongest bursts of collaborative behavior come from those wishing to appear more like their more-elite peers.

Status-aspiring actors may also exert agency by making "ceremonial expenditures" as originally discussed by Meyer and Rowan (1977), who noted the increase in status that can arise for a university that hires a Nobel Prize winner. Like status-aspirational pricing, hiring a Nobel laureate is possible only for institutions that fit the criteria implied by our moderating hypotheses: they are core, rather than peripheral, market actors, and their stakeholders are not expected to bear unfair costs. Unlike status-aspirational pricing, however, hiring an academic superstar rarely happens quickly. New research on the possible effect of status loss on ceremonial hiring would therefore need to consider an important, if hard-to-measure, moderator, namely, whether the organization suffering a status decline has the staying power or collective patience to wait for an external savior (cf. Khurana, 2004).

We also see two implications for future investigations in performance feedback theory. First, consistent with schools' setting tuition merely to improve how they are perceived, we see valuable opportunities for theorists of organizational learning to model what might be termed cosmetic, as opposed to material, responses to unfulfilled aspirations. Most of the earlier work in this tradition has focused on material outcomes, such as R&D intensity and strategic change (Greve, 2003). In contrast, we see interesting possibilities for modeling the hazard of symbolic reactions (e.g., securing celebrities' endorsements) as a function of slips in performance. Second, given our dependent

variable, an obvious suggestion for subsequent research is to consider whether the loss of market share (potentially similar to loss in prestige) could trigger status-aspirational pricing. Shipilov, Li, and Greve (2011) used relative market share as a predictor of changes in partnering behavior, but pricing changes could also be modeled. If a company loses market share, we generally expect it to lower prices to recover customers or to seek orthogonal market niches. But there are some markets—particularly those described by White (1981: 540–541) and Jensen, Kim, and Kim (2011: 93–94)—in which market share is positively associated with producers' status. In such domains, there is a natural link to performance feedback theory arising from our approach: are there conditions in which producers strategically raise price in response to shortfalls in market share?

Turning to administrative research on ranking systems (e.g., Elsbach and Kramer, 1996; Sauder, 2008), our moderating effects lead us to anticipate that new research in this vein will profit from focusing on interorganizational sources of variation in susceptibility to third-party arbiters (Sauder and Fine, 2008). Sauder (2008) offered a rich account of the process by which *USN* adroitly inserted itself as the key arbiter of perceived quality in U.S. legal education. *USN* first won over students and employers, and from that beachhead brought about increased homogeneity across law schools, which felt obliged to bend to its criteria. Sauder's analysis also indicated that law schools with less experienced deans were most susceptible to the influence of *USN*. Building on this theme, a valuable next step is to bring a network-analytic perspective to bear on the historical question of which schools succumbed and which ones resisted *USN*'s pressures toward conformity. For instance, law schools whose structurally equivalent peers initially posted fast improvements on metrics imposed by *USN* may have been those that conformed most quickly and tightly. Alternatively, schools encircled by conformist peers may have been those most prone to resist and delay in an effort to appear authentic.

Limitations and Scope Conditions

Additional possibilities for new research arise from considering two limitations of our analyses and four scope conditions on the generalizability of our hypotheses. First, in focusing simply on whether a focal school was discernibly below its reference point, we did not consider potential moderating effects of different rates of status decline. Descent (and ascent) can be rapid or gradual, and if it is gradual, the sliding organization may fail to respond—much like a slowly boiled frog. New research could consider how different rates of status change exert indirect, or even direct, effects on future organizational behavior.

Second, we may not have fully captured the motivations of the organizational decision makers in our setting. In particular, we did not address whether the price signal may also be intended for an internal audience to trigger a self-fulfilling prophecy among employees. As Klein and Leffler (1981: 634) pointed out, "price not only influences buyers' expectations but also influences producers' incentives." Although we cannot address this possibility with our current data, it is worth examining whether a price setter (e.g., provost) may be pursuing intra-organizational changes as he or she prices up: in this scenario, the goal is to push faculty and administrators to live up to the new price. Seeking to better understand if and when the Chivas Regal strategy is pursued to motivate

internal stakeholders is an interesting possibility for future work, especially research taking a qualitative approach.

Moreover, some institutional features of higher education are certainly not present across all markets, so status-aspirational pricing will not occur universally. In addition to the two conditions implied by our moderating hypotheses—the focal firm must be viewed as a credible member of the core of its market, and it cannot violate norms of fairness when it determines its price—four further boundary conditions are required. When these conditions are met by at least a subset of the buyers and producers in a given setting, we expect producers in that setting to increase price in reaction to status loss. Two conditions relate to buyers, and two concern a focal producer.

First, buyers face uncertainty about the quality of the producer's good. Without this uncertainty, price is a rigid reflection of the good's known or observable attributes, and a producer therefore cannot use it as a strategic, aspirational device. In contrast, goods often marked by sufficient uncertainty for status-aspirational pricing include experience goods (Nelson, 1970), credence goods (cf. Rössel and Beckert, 2013) like a fair-trade food whose conformity to sustainability ideals buyers must blindly trust, and Veblen goods, whose ability to confer status on buyers is known only after consumption. Uncertainty creates degrees of freedom in setting prices. Producers of such goods not only have more information about their goods than do their buyers (Akerlof, 1970), but their products or services are also marked by a second-order type of uncertainty in which the dimensions that matter for buyers' assessments of quality are multiple and contested, rather than singular and taken for granted (Rössel and Beckert, 2013). The producer thus has the flexibility to set its price with enhanced status as its aim: the price itself can enter the mix of quality dimensions that are up for grabs, and—in the producer's decision-making process—can be seen as a route to status gain.

Second, buyers have historically viewed high prices as positive signals of quality and/or status. Buyers may intuitively understand that only a truly high-quality producer can sustain a high price without losing customers (Bagwell and Riordan, 1991), or they may see their purchase of more expensive goods as a step to advancing in status themselves (Lichtenstein, Ridgway, and Netemeyer, 1993). Not all buyers must meet this second scope condition; only a subset must. Many buyers may have long been strongly averse to high prices, while some sophisticated consumers may identify price hikes as a placebo (cf. Pettit, Sawa, and Sawa, 1985; Cohen and Chakravarti, 1990). What is important is that enough buyers have, over a sustained period of time, viewed high prices as positive signals of quality and/or status. When this is the case, a producer can then realistically believe that raising its price is a viable strategy.

Third, organizational status is mutable and is an important objective for the producer. Not all status hierarchies meet these criteria. Mills (1951: 239–240) distinguished between two hypothetical status systems: one in which "the exact volume and types of deference expected between any two individuals are always known, expected, and given" and a second "in which prestige is highly unstable and ambivalent." For our hypotheses to apply, the focal producer and its peers must occupy a status hierarchy closer to the second type. This means that the producer is not so secure in its role (Phillips and Zuckerman, 2001) that slips in status never occur. Nor does that producer face cultural norms that punish status falls so sharply that recovery is untenable

(Bothner, Podolny, and Smith, 2011). In addition, the producer must significantly value its status beyond the goal of sustained financial performance (Shipilov and Li, 2008), and it must do so in the particular status hierarchy in which the loss occurs. For instance, a business school that fares well in several published rankings (none of which enjoy the hegemony of *USN* for undergraduate intuitions) may value its status but still discount a sharp loss in one of the rankings (cf. Elsbach and Kramer, 1996). For that business school, the rank decline is a dismissible case of status inconsistency, not cause for alarm. Where we observe status-aspirational pricing, we will necessarily see producers for which status is both a salient goal in the hierarchy in question and subject to (reasonable and recoverable) shifts across time.

Fourth, the producer perceives some form of safety net that lowers the perceived risk of aspirational pricing.[12] This condition is informed by one of the unique aspects of our empirical setting: the presence of a third-party subsidizer, the U.S. government. Government-subsidized financial aid allows colleges and universities to engage in price discrimination on the basis of students' socioeconomic status. In our empirical setting, this is fully legal and sustainable because admitted students cannot resell their spot (to richer, less-deserving applicants). Clearly, this makes it easier to raise prices after a slip in rank. Nonetheless, we do not see the presence of an active subsidizer as necessary for our theory to apply in other industries; substitutes exist for this role in other contexts. Such substitutes could include, at the most basic level, access to credit or an appetite for risk. Imagine, for example, a solo-practitioner lawyer who experiences a status drop: two close rivals recently secured new, prestigious clients, bringing them a halo he or she now lacks. Suppose also that his or her response is to raise hourly rates for reasons consistent with our theory: a higher price makes the lawyer seem closer to competitors in the local status hierarchy. Our hypothetical lawyer lacks government support but relies on two substitutes: willingness to risk short-term drops in demand to send the right, price-related message and access to capital should things go wrong. Though our initial boundary condition on producer credibility is similar conceptually, credibility is analytically distinct from the perception of a buffer against risk. When such a buffer exists, status-aspirational pricing is more likely to occur.

Settings for Future Research

For organizations that meet the scope conditions we have described, two industries are especially promising sites for replicating and extending our results: wine and high-end restaurants. Similar to other settings that are also candidates for future study—art, cosmetics, consulting firms, and luxury goods—these industries contain producers with the credibility and latitude to price in ways intended to counteract shortcomings in status. Wineries and elite restaurants are particularly appealing, due to the precise time-varying data maintained on organizational status and pricing.

Starting with wine, perhaps the most straightforward way to replicate our findings is to assemble a panel similar to that used by Benjamin and Podolny

[12] We are grateful to an anonymous reviewer for bringing into focus the issues related to this boundary condition.

(1999) and then take three analytical steps: add winery fixed effects, construct indicator variables capturing drops in status below a reference point, and examine interactions with measures of credibility and of close competitors' lagged prices. Benjamin and Podolny's measure of reputation could inform a covariate capturing credibility, and computing rivals' average prior prices is feasible in light of the relational data available. Our falsifiable prediction is a positive main effect of status loss on future price. We believe this is likely in light of the well-documented connection between price and perceived quality in wine (cf. Plassman et al., 2008). We also expect a set of moderating effects similar to what we have observed. We expect a certain subset in a panel of wineries will exhibit patterns consistent with our theory. If we are correct, then this core question warrants close attention in future work: what are the distinctive features of organizations in this section of the wine market?

Snyder and Cotter (1998) offered evidence largely in keeping with our theory in their analysis of French restaurants gaining and losing Michelin stars. They revealed that restaurants dramatically increased their prices in the two years prior to receiving a Michelin star. This effect is consistent with our main prediction: when shy of an important status marker, a focal organization will raise prices in hopes of ascending. In addition, although Snyder and Cotter found that restaurants demoted from three stars lowered their prices commensurately, the experience of two-star restaurants experiencing a loss in status is provocative: after suffering demotion to just one star, these restaurants—slipping from the elite core of this market but still credible members of it—raised their prices. An interesting and straightforward next step involves identifying when these downwardly mobile restaurants raise prices most sharply in response.

Contrary to what prior explorations of status and its effects imply, our account shows that organizations are not always passive recipients of their status and its consequences. Quite the opposite: in settings in which status is both alterable and a primary organizational goal—like higher education, wine, and fine restaurants—firms will actively take steps to change their position in a status hierarchy. The story of Chivas Regal whiskey provided an apposite analogy for the types of contexts in which organizations will actively attempt to signal higher status, as well as the means by which they will do so. When organizations conclude that passively accepting a lower-than-aspiration level of status is intolerable, they will raise their prices as a means of dynamically responding to a loss. We expect future research on status to more seriously consider such agentic responses from both firms and individuals when examining shifts in status hierarchies over time.

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